

TriZOD — Final Pipeline & Re-Referenced Dataset

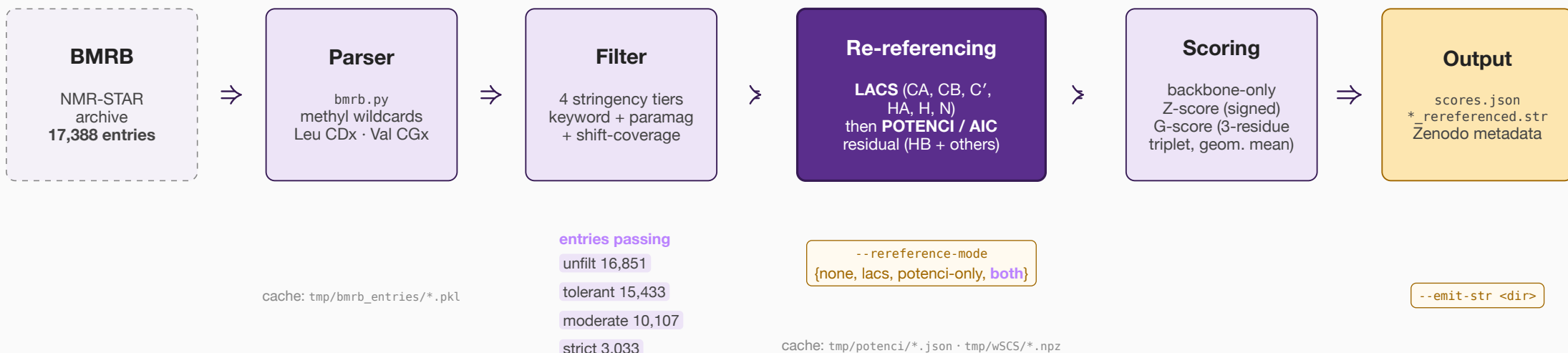
Finalized pipeline · α -synuclein case study · chemical-shift perturbations

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6 May 2026

TUM · TriZOD project meeting

TriZOD pipeline at a glance

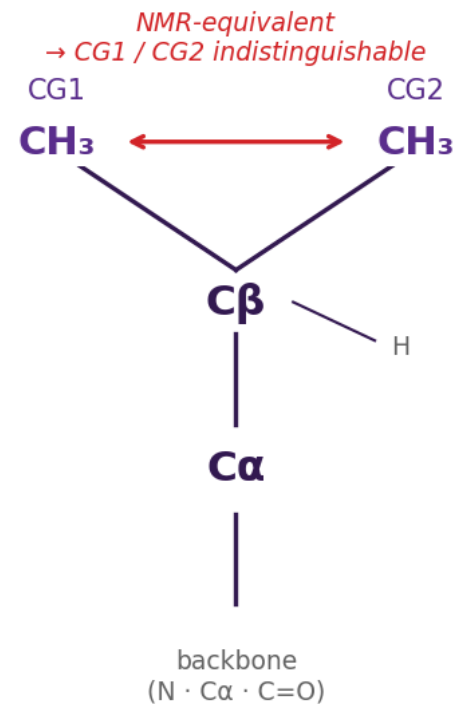


Next steps:

- exclude multi-molecule (bound) entries that distort G-scores
- per-sequence representative pick (best conditions → median G-score)
- mmseqs2 sequence clustering for ML train/val/test split

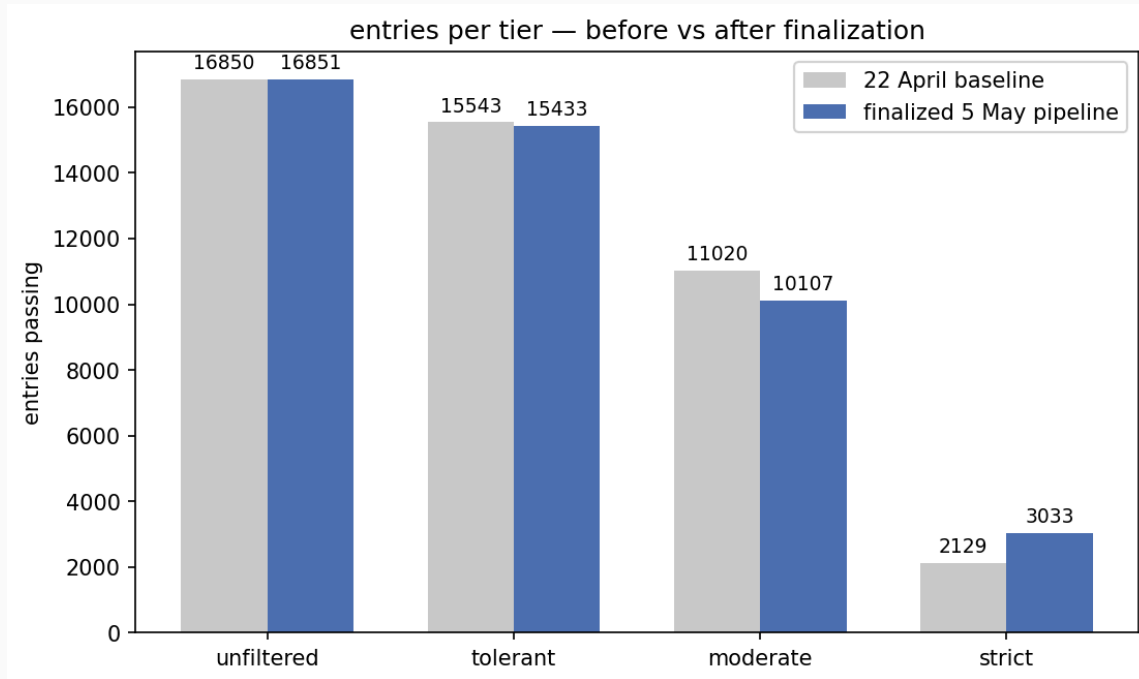
Implementation update

- Methyl wildcards in the parser (Leu CD1/CD2 → CD_x, Val CG1/CG2 → CG_x)
- LACS pre-correction baked into the scoring pipeline
- New flag: `--rereference-mode {none, lacs, potenci-only, both}` (default both)
- New flag: `--emit-str <dir>` writes one re-referenced NMR-STAR per scored entity



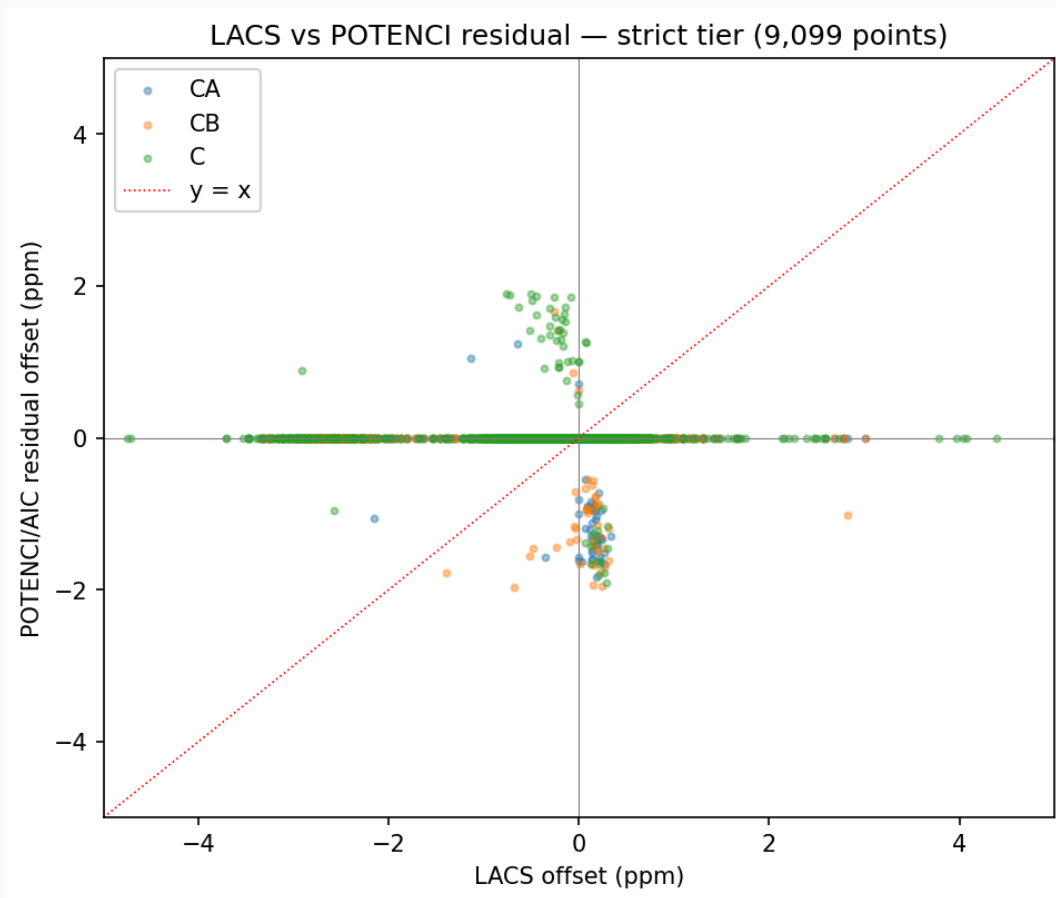
Valine side chain — geminal methyls

Filter improvements and per-tier dataset deltas



Removed denaturant false-positives, fixed the solid-state regex, added a paramagnetic-sample filter, relaxed `min-backbone-shift-types 5→4` in strict, and broadened the Celsius heuristic.

LACS vs POTENCI residual capture

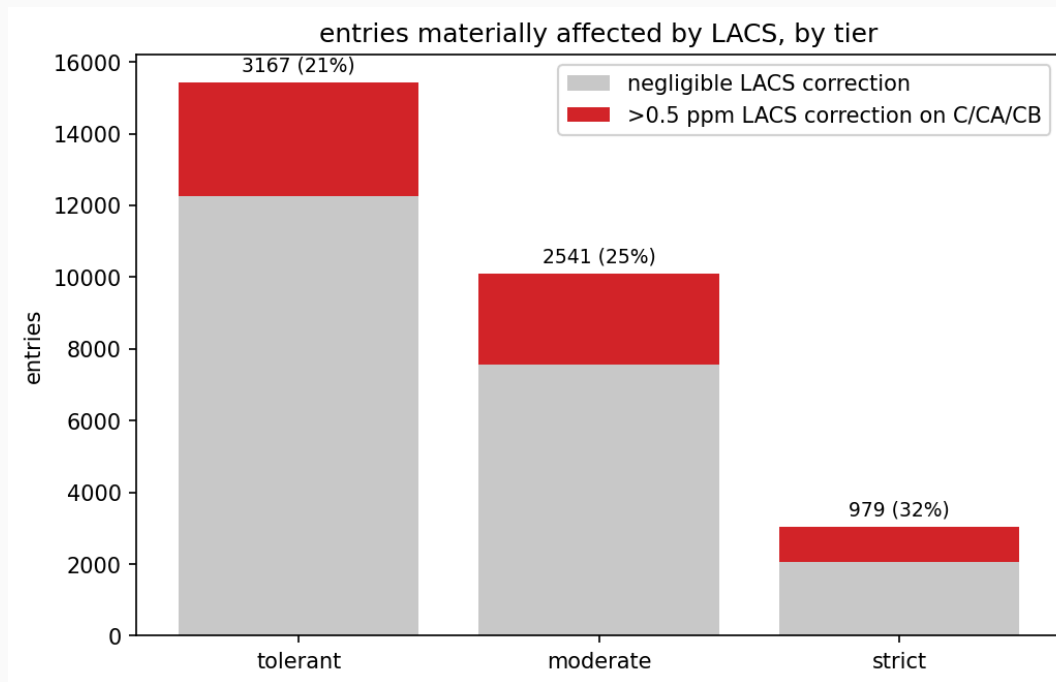


Each dot is **one entry × one atom** (CA / CB / C, strict tier).

LACS catches the large systematic referencing bias.

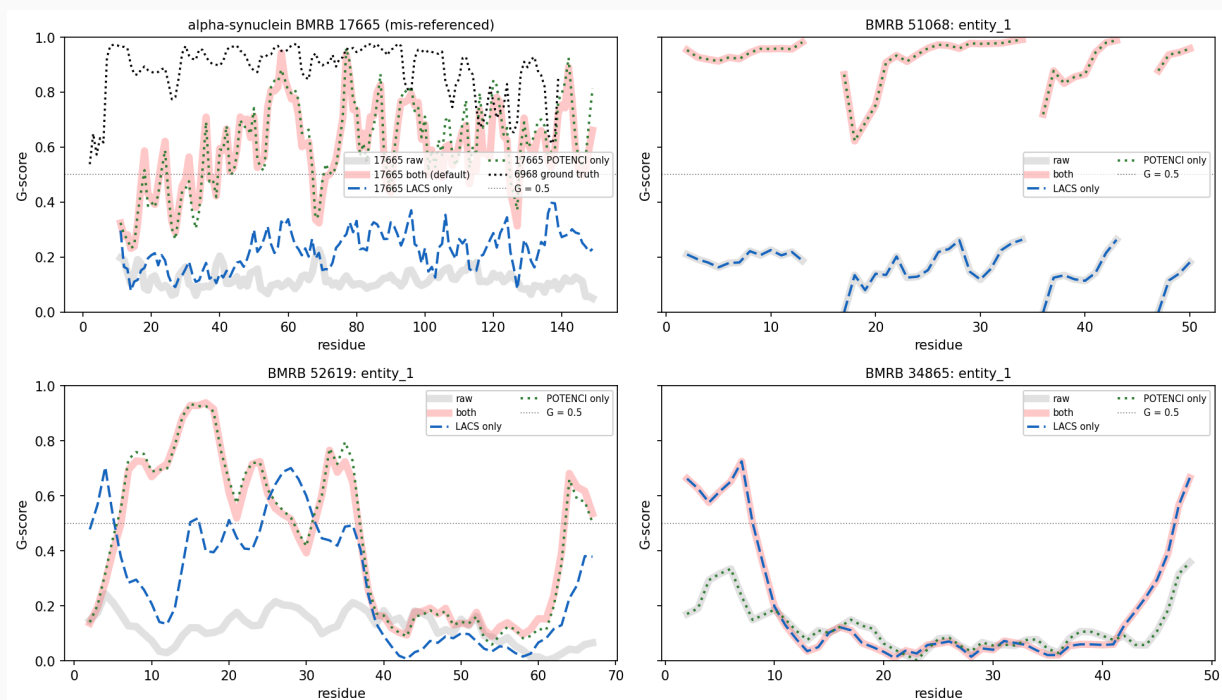
POTENCI/AIC residual mops up the remainder — most points land near the origin once LACS has done its job.

Entries affected by LACS



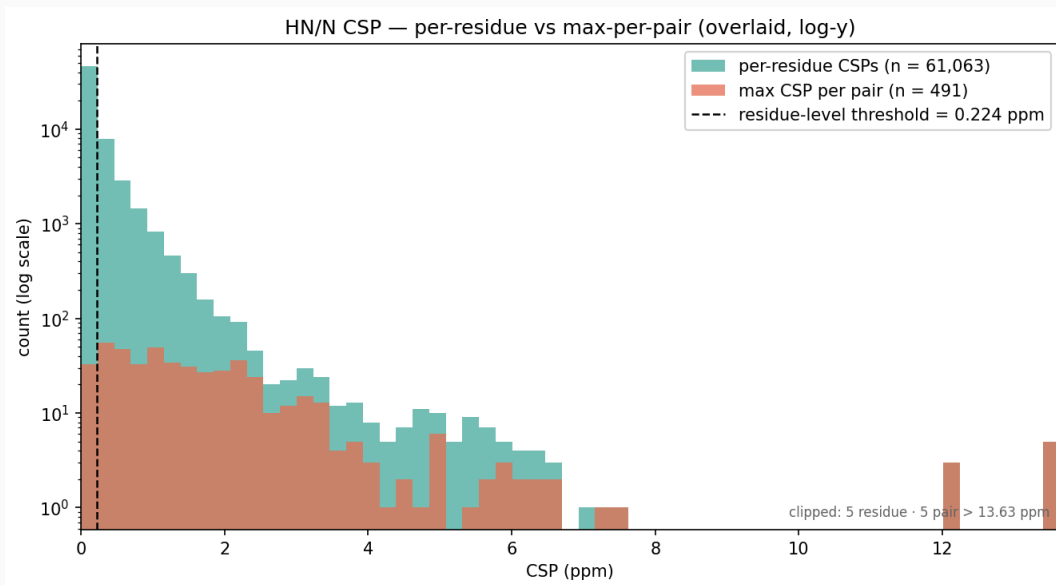
Red = entries with $|\text{LACS offset}| > 0.5$ ppm on at least one of C/CA/CB. Re-referencing meaningfully changes the input shifts for **21%** (tolerant), **25%** (moderate), **32%** (strict).

α -synuclein and G-score flippers — POTENCI-dominant vs LACS-dominant

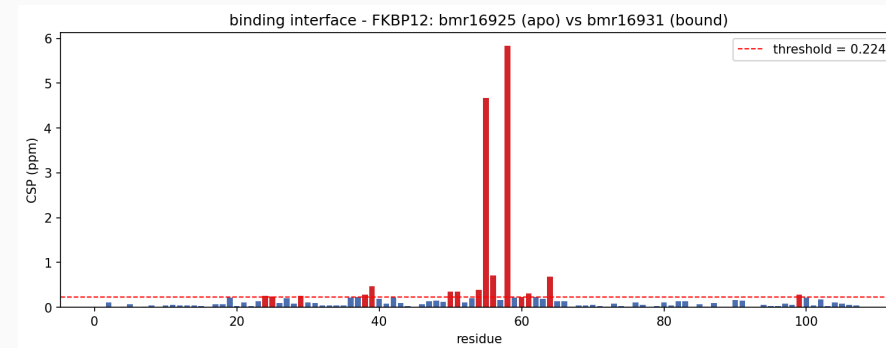


Panel A · BMRB **17665** (α Syn, mis-referenced) raw vs re-referenced + BMRB **6968** (α Syn ground truth). Panels B/C · top POTENCI-dominant flippers BMRB **51068**, **52619** (POTENCI/AIC catches the offset; LACS quiet). Panel D · BMRB **34865** **LACS-dominant** (LACS catches the offset; POTENCI/AIC's AIC threshold rejects it). Gaps from $k = 0$ triplets (insufficient comparable shifts $\rightarrow$ G-score is NaN). α Syn LACS offset CA / CB = +2.82 ppm (matches Reid's TALOS-N reading).

Chemical-shift perturbation distribution



Teal: 61,063 per-residue HN/N CSPs · **orange:** one max CSP per pair (n = 491/581). Threshold **0.224 ppm** = trimmed-mean+SD on the bottom 90%. 90 pairs dropped — no overlapping HN+N coverage.



FKBP12 apo (bmr16925) vs bound (bmr16931). Two binding-pocket residues (55, 58) far above threshold.

$$\text{CSP} = \sqrt{(\Delta\delta_H)^2 + (\Delta\delta_N/\alpha)^2}, \quad \alpha = 5$$

$$\Delta\delta_X = \delta_X^{\text{bound}} - \delta_X^{\text{free}} \cdot \alpha = 5 \text{ rescales } ^{15}\text{N} \text{ to } ^1\text{H} \text{ ppm}$$

Released artefacts

- `data/release/<tier>/scores.json` — per-residue Z/G + LACS + POTENCI offsets
- `data/release/<tier>/str/bmr*_*_rereferenced.str` — re-referenced NMR-STAR
- `.zenodo.json` + `CITATION.cff` — DOI on first tagged release

Per-tier counts after the rerun:

Tier	Entries	Coverage
strict	3,033	17.00%
moderate	10,107	56.64%
tolerant	15,433	86.49%
unfiltered	16,851	94.45%

Next steps

- Finalize dataset for ML
- Push deposit to Zenodo (DOI placeholder until first tag)
- Iterate CSP extend to multi-atom $\Delta\omega_{\text{RMS}}$ (Schumann/Williamson 2008)
- PDB structure matching idea